NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2024
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED	MATHEMATICS	M.M. 72
REVISION ASSINGMENT # 05	(PROBABILITY)	TIME : 60 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. A bag contains 3 red, 2 green and 2 white balls. Two balls are drawn at random and none of them is found to be a green ball, then the chance that both are red is equal to -

(A) $\frac{1}{10}$ (B) $\frac{2}{10}$ (C) $\frac{3}{10}$ (D) $\frac{4}{10}$

2. If A & B are independent events such that $P\left(A \cap \overline{B}\right) = \frac{1}{3}$ & $P\left(A \cup B\right) = \frac{11}{15}$, then $P(A \cap B)$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{5}{11}$ (C) $\frac{2}{9}$ (D) $\frac{7}{9}$

3. If 'P' is probability of an event E of a random experiment and $\{P\} = P$, then (where $\{.\}$ denotes fractional part function) -

(A) E must be a sure event

(B) E can't be a sure event

(C) $\{P(\overline{E})\} = P(\overline{E})$ for all possible values of P(E)

(D) $\{P(\overline{E})\} \neq P(\overline{E})$ for all possible values of P(E)

4. A person has to catch a train. To catch train, from his home he can take a taxi or take rickshaw or walk by foot with respective probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$.

Probability of him catching train when he takes rickshaw from his home is half that of when the takes the taxi and probability of catching the train when he walked by foot is $\frac{1}{4}^{th}$ that of when he takes rickshaw.

He finally reached the train, the probability he walked by his foot to catch the train, is

(A) $\frac{1}{33}$ (B) $\frac{2}{33}$ (C) $\frac{7}{33}$ (D) $\frac{13}{33}$

5. A fair die is thrown repeatedly. If '3' appears on first three throws, then the probability of '4' appearing on the fourth throw, is-

(A) $\frac{1}{216}$ (B) $\frac{1}{36}$ (C) $\frac{1}{1296}$ (D) $\frac{1}{6}$

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6. If $P(A)=\frac{3}{4}, P(A\cap B\cap \bar{C})=\frac{1}{3}$, $P(A\cap \bar{B}\cap \bar{C})=\frac{1}{3}$. Then P(A∩C)=

(A) $\frac{1}{6}$ (B) $\frac{1}{15}$ (C) $\frac{1}{12}$ (D) $\frac{1}{9}$

7. A biased die is such that probability of obtaining face numbered i is proportional to i. If die is rolled twice and faces 'a' and 'b' turn up on first and second turn respectively, then probability that a is even and b is odd is-

(A) $\frac{12}{49}$ (B) $\frac{9}{49}$ (C) $\frac{17}{49}$ (D) $\frac{8}{49}$

8. A bag contains 5 white and 3 black balls. 4 balls are drawn one by one, their color noted and replaced in the bag. Probability that white and black balls are drawn alternately is-

(A) $\frac{15^2}{2^{10}}$ (B) $\frac{15^2}{2^{11}}$ (C) $\frac{15^2}{2^{12}}$ (D) $\frac{15^2}{2^{13}}$

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

9. In an experiment A & B events are such that $P(\bar{A})=\frac{6}{11}; P(B)=\frac{5}{22}, P(A\cap B)=\frac{1}{11}$, then -

(A) $P(A\cup B)=\frac{15}{22}$ (B) $P(\bar{A}\cap\bar{B})=\frac{9}{22}$ (C) $P(\bar{B}\cap A)=\frac{4}{11}$ (D) $P(\bar{A}\cup\bar{B})=\frac{10}{11}$

10. If X and Y are independent then P(X ∩ Y) = P(X). P(Y). Let A & B, B & C, C & A are three pairs of independent events such that $P(A)=\frac{1}{2}, P(B)=\frac{1}{3}, P(C)=\frac{1}{6}$ then-

(A) $P(\bar{A}\cup\bar{B})=\frac{5}{6}$ (B) $P(B\cap\bar{C})=\frac{5}{18}$

(C) $P(\bar{A}\cap\bar{B})=\frac{1}{3}$ (D) $P(\bar{A}\cap B)+P(\bar{B}\cap C)+P(\bar{C}\cap A)=\frac{25}{36}$

11. Let A, B and C be three independent events of an experiment such that $P(A)=\frac{1}{3}$, $P(B)=\frac{1}{2}$ and $P(C)=\frac{1}{4}$. Which of the following statement(s) is/are correct ?

(A) Probability that atleast one of three events happening is 3/4.

(B) Probability of exactly two of these happening is $\frac{1}{4}$.

(C) Probability of atleast two of them happening is $\frac{7}{24}$.

(D) Probability of atmost two of them happening is $\frac{23}{24}$.

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Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 12 to 13

Three players A, B, C alternately throw a die in the order. The first player to throw a 6 will be the winner. A's die is fair where as B and C throw a biased die with probability of throwing a 6 being p_1 and p_2 respectively.

12. Ratio of probability of winning of B to probability of winning of C is -

(A) $\frac{1-p_1}{p_1p_2}$ (B) $\frac{p_1}{1-p_1}$ (C) $\frac{p_1}{p_2(1-p_1)}$ (D) $\frac{1-p_1}{p_1}$

13. Ratio of p_1 and p_2 so that the game becomes equally likely for all the three players is -

(A) 5 : 4 (B) 1 : 2 (C) 4 : 5 (D) 2 : 1

Paragraph for Question 14 & 15

Three sample papers x, y, z are available in the market. A student who purchase only one sample paper, purchases x, y, z with the probabilities $\frac{1}{7},\frac{1}{3},\frac{11}{21}$ respectively. Student will pass with the probabilities $\frac{1}{2},\frac{1}{3},\frac{1}{6}$ if he study sample papers x, y, z respectively.

14. Probability that student will pass in the examination is -

(A) $\frac{17}{63}$ (B) $\frac{43}{63}$ (C) $\frac{46}{63}$ (D) $\frac{23}{63}$

15. If it is known that student is fail, then probability that he studied sample paper 'y', is -

(A) $\frac{9}{92}$ (B) $\frac{55}{92}$ (C) $\frac{7}{23}$ (D) $\frac{2}{9}$

SECTION–II

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. If two numbers are randomly chosen from the set of first 30 natural numbers, then probability that atleast one of them is prime is given by $\frac{a}{b}$, where a & b are coprime, then value of $\frac{b-a}{19}$ will be

2. Bag I contains 4 white and 3 black balls and another bag II contains 5 white and 2 black balls. Two balls are drawn randomly from the first bag and put in the second bag and then a ball is drawn from the bag II. The probability that the ball drawn from the second bag is white is expressed as $\left(\frac{p}{q}\right)$ where p, q ∈ N. Then the minimum value of (p + q) is

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3. Humpty plays three matches against Dumpty. In any match the probabilities of Humpty getting 0, 2 and 3 points are 0.2, 0.3 and 0.5 respectively. If probability that Humpty getting atmost 6 points is $\frac{p}{200}$, then then sum of the digits of 'p' is

4. A hunter's chance of shooting an animal at a distance r is $\frac{a^2}{r^2}$ $(r > a)$. He fires when $r = 2a$ & if he misses he reloads & fires when r = 3a, 4a,..... If he misses at a distance '50a', the animal escapes. Find the probability that the anmial escapes.

SECTION–IV

Matrix Match Type (One or More than one option correct)

For each entry in **Column-I** , +2 If only the bubble(s) corresponding to all the correct matche(es) is (are) darkened, 0 In none of the bubbles is darkened, –1 In all other cases

1.

	Column-I		**Column-II**
(A)	A bag contains two red, two green and a certain number of black balls. Two balls are drawn at random and the probability that they are of different colour is $\frac{5}{7}$, then number of black balls is equal to	(P)	4
(B)	Consider two independent events A & B where $P(A)=\frac{1}{3}$, $P(B)=\frac{1}{4}$ & $P(A/\bar{B})=\frac{a}{6}$, then 'a' is equal	(Q)	2
(C)	Minimum number of times a dice must be rolled so that the chance of getting one atleast once exceeds 0.5 is equal to	(R)	3
(D)	Odds in favour of getting exactly 3 heads in the simultaneous throw of 8 coins is "m to 25" then 'm' is equal to	(S)	odd
		(T)	prime

Space for Rough Work

REVISION ASSINGMENT # 04 **P & C, BT** **MATHEMATICS**

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	B	C	B	C	B	C	A	B,C	A,C	A,B,C,D
	Q.	11	12	13	14						
	A.	A,B,D	D	C	A						
SECTION-II	Q.	1	2	3	4						
	A.	5.00	0.00	196.00	8.00						

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